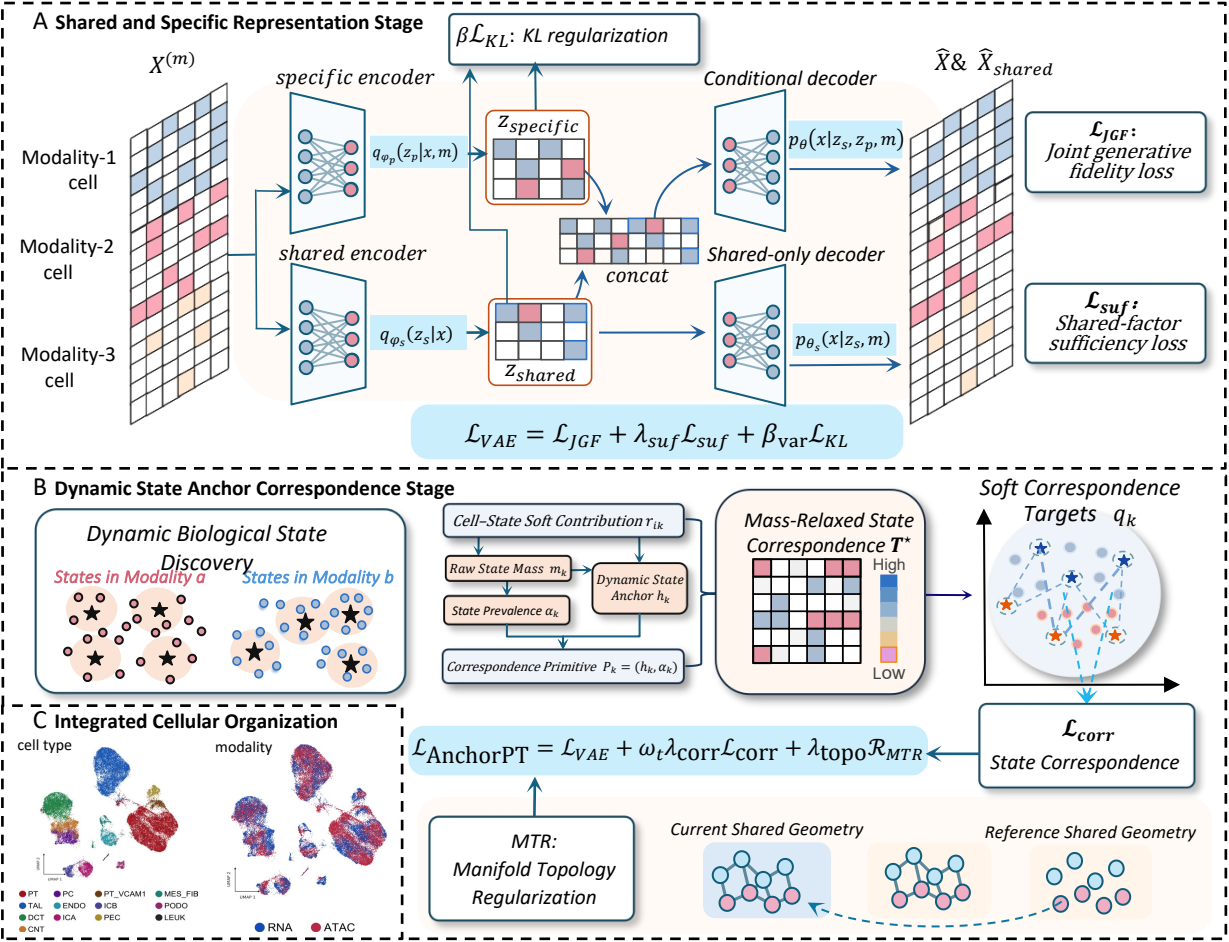

A Shared and Specific Representation Stage
$X^{(m)}$
Modality-1 cell
Modality-2 cell
Modality-3 cell
specific encoder
shared encoder
$q_{\varphi_p}(z_p|x,m)$
$q_{\varphi_s}(z_s|x)$
$\beta\mathcal{L}_{KL}$: KL regularization
$z_{specific}$
z_{shared}
concat
Conditional decoder
Shared-only decoder
$p_\theta(x|z_s,z_p,m)$
$p_{\theta_s}(x|z_s,m)$
$\widehat{X}$& $\widehat{X}_{shared}$
$\mathcal{L}_{JGF}$: Joint generative fidelity loss
$\mathcal{L}_{suf}$: Shared-factor sufficiency loss
$\mathcal{L}_{VAE} = \mathcal{L}_{JGF} + \lambda_{suf}\mathcal{L}_{suf} + \beta_{\text{var}}\mathcal{L}_{KL}$
B Dynamic State Anchor Correspondence Stage
Dynamic Biological State Discovery
States in Modality a
States in Modality b
Cell–State Soft Contribution r_{ik}
Raw State Mass m_k
Dynamic State Anchor h_k
State Prevalence α_k
Correspondence Primitive $P_k = (h_k, \alpha_k)$
Mass-Relaxed State Correspondence $T^\star$
High
Low
Soft Correspondence Targets q_k
$\mathcal{L}_{corr}$ State Correspondence
C Integrated Cellular Organization
cell type
modality
PT
PC
PT_VCAM1
MES_FIB
TAL
ENDO
ICB
PODO
DCT
ICA
PEC
LEUK
CNT
RNA
ATAC
$\mathcal{L}_{\text{AnchorPT}} = \mathcal{L}_{VAE} + \omega_t\lambda_{\text{corr}}\mathcal{L}_{\text{corr}} + \lambda_{\text{topo}}\mathcal{R}_{MTR}$
MTR: Manifold Topology Regularization
Current Shared Geometry
Reference Shared Geometry